

Lab 3 – Deep Packet Inspection with Zeek

➤ Learning Objectives

By the end of this lab, students should be able to:

- Understand how Zeek monitors and logs network activity
 - Analyze network traffic through Zeek logs
 - Identify communication patterns between hosts
 - Detect suspicious behaviors without relying on signatures
 - Develop a network analysis mindset
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➤ Prerequisites

Students must have:

- Completed previous labs using **Suricata**
 - **Zeek** installed and operational on a **Linux system**
 - Basic networking knowledge (TCP, IP, DNS, HTTP)
 - Access to a **PCAP file**
 - Reference documentation:
<https://docs.zeek.org/>
<https://www.malware-traffic-analysis.net/>
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➤ Lab Scenario

You are a **network security analyst**.

Unlike previous labs, you are not given alerts.

Instead, you must **analyze raw network activity logs** to understand what is happening.

Your goal is to:

- Identify communication patterns
- Detect anomalies
- Infer potential security incidents

Part 1: Generating Zeek Logs

Mission 1: Process a PCAP File

Zeek does not generate alerts like an IDS. It generates **structured logs**.

Tasks

- Run Zeek on a given PCAP file
- Identify the generated log files

Deliverables

- List of generated log files
- Number of logs created

Mission 2: Understand Log Structure

Zeek produces multiple logs, each representing a different aspect of network activity.

Tasks

Identify and understand the purpose of the following logs:

- conn.log
- dns.log
- http.log

Deliverables

For each log:

- What type of information it contains
- Why it is useful

Part 2: Network Behavior Analysis

Mission 3: Analyze Connections

Focus on **conn.log**

Tasks

- Identify:
 - Source IPs
 - Destination IPs
 - Connection durations
- Detect unusual patterns:
 - Repeated connections
 - Short-lived connections
 - High number of connections

Deliverables

- At least 2 suspicious connection patterns
- Explanation of why they are suspicious

Mission 4: Analyze DNS Activity

Now focus on **dns.log**

Tasks

- Identify domains being queried
- Detect:
 - Unusual or random-looking domains
 - High frequency of DNS requests
- Identify any potential suspicious domains

Deliverables

- List of suspicious domains
- Justification

Mission 5: Analyze HTTP Traffic

This time focus on **http.log**

Tasks

- Identify:
 - Requested URLs
 - HTTP methods
 - User agents
- Detect:
 - Unusual requests
 - Suspicious user agents
 - Repeated requests

Deliverables

- Example of suspicious HTTP activity
- Explanation

Part 3 : Correlation Analysis

Mission 6: Identify Suspicious Behavior

Using all logs combined:

Tasks

- Correlate information from:
 - conn.log
 - dns.log
 - http.log
- Identify at least **one potential incident scenario**

Deliverables

- Description of the scenario (5–10 lines)
- Supporting evidence from logs

Tips & Best Practices

- Zeek **does not detect**, it **describes behavior**.
 - Focus on **patterns**, not isolated events.
 - Look for:
 - repetition
 - anomalies
 - unusual destinations
 - Always ask: *“Does this behavior make sense?”*
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Reflection Questions

- What is the main difference between Zeek and an IDS like Suricata?
- Why is visibility important in network security?
- How can suspicious behavior be detected without predefined rules?
- What are the limitations of log-based analysis?
- Why is correlation between logs important?

“Suricata tells you something is wrong. Zeek tells you why”